



**AIR UNIVERSITY**  
**DEPARTMENT OF MECHATRONICS ENGINEERING**  
**SEMESTER: SPRING 2026**  
**Computer Programming**  
**CE-112**

**Assignment-2**

**Instructor: Umer Farooq**

**Due Date: 07<sup>th</sup> May 2026**

**Soft Copy Time: 10:00 PM**

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| <b>Question No</b> | <b>Write Status<br/>(Complete/Not complete and issue that you faced) (Compulsory, Question will be checked on bases of this index)</b> | <b>Remarks<br/>Must be filled for every question</b> |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
| <b>1</b>           | Completed                                                                                                                              | Accurately finds minimum and maximum array values    |
| <b>2</b>           | Completed                                                                                                                              | Computes standard deviation using the math library   |
| <b>3</b>           | Completed                                                                                                                              | Computes 3x3 matrix multiplication successfully      |
| <b>4</b>           | Completed                                                                                                                              | Successfully parses strings for hidden characters    |
| <b>5</b>           | Completed                                                                                                                              | Calculates mean, median, and mode correctly          |

Q1) Write and test the following function that returns through its reference parameters both the maximum and the minimum values stored in an array: void getExtremes(float& min, float& max, float a[], int n);

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### Extremes

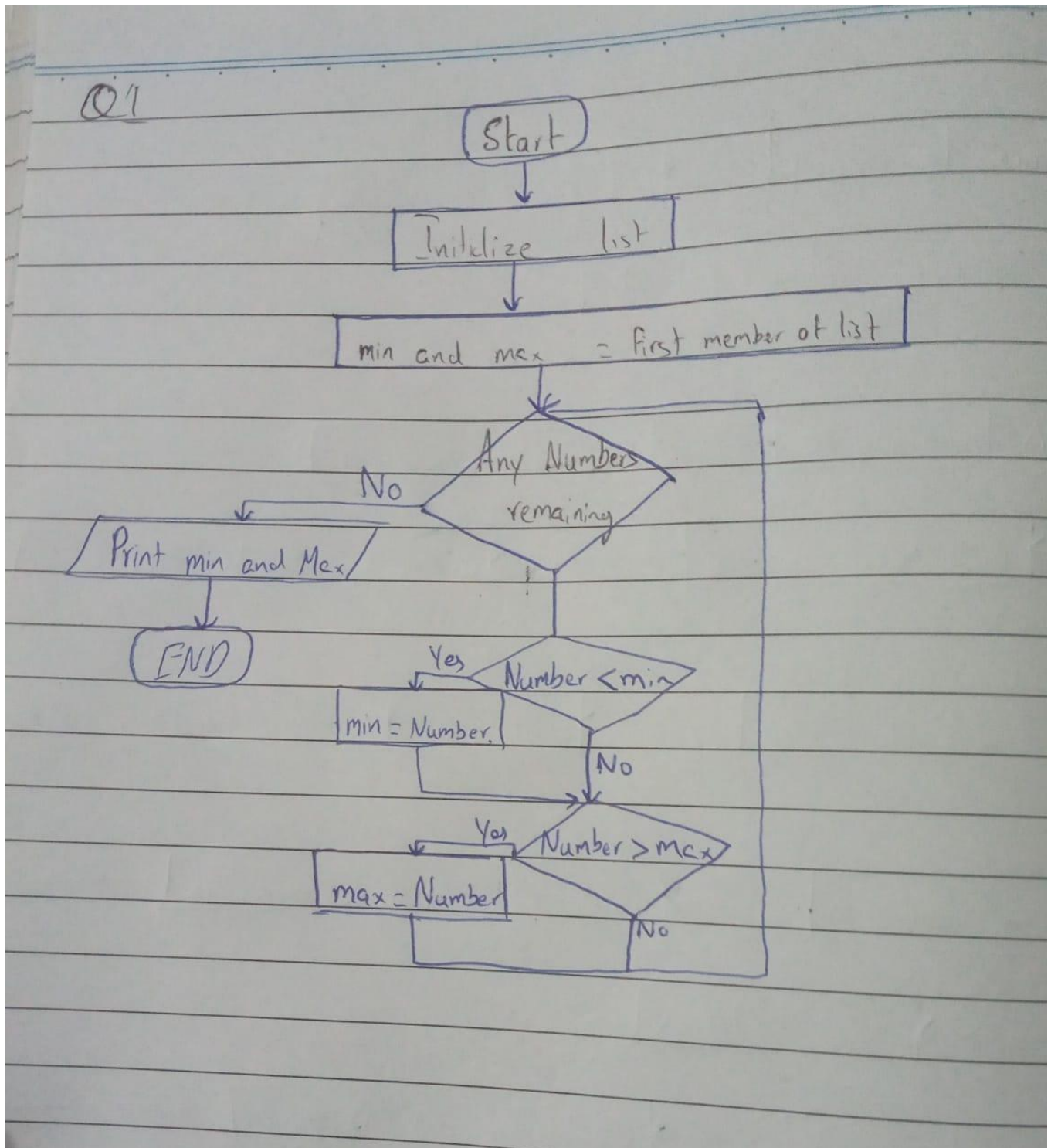
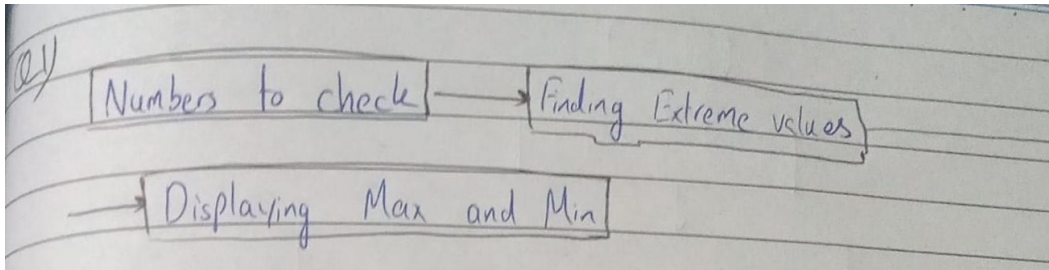
```
#include <iostream>
using namespace std;

int main() {
    void getExtremes(float& min, float& max, float a[], int n) {
        // & signs to pass values by reference
        min = a[0]; max = a[0] // Allocate first value to max and min

        for(int i=0; i<n; i++) {
            if(a[i] > max)
                max = a[i]; // if current number is greater than max replace it
                               // with max
            if(a[i] < min)
                min = a[i]; } }

    int main() {
        float min, max, n;
        float number[] = {56, 77, 1, 23, 22};
        n = sizeof(number) / sizeof(number[0]);
        getExtremes(min, max, number, n);

        cout << "The minimum in the array is: " << min << "\n The maximum
        is: " << max;
    }
}
```



```
The minimum in the array is: 1
The maximum is: 77
PS C:\Users\ESHOP\Documents\Semester_2\CP\CP_A2\Extremes>
```

Q2) Write and test the following function:

double stdev(double x[], int n);

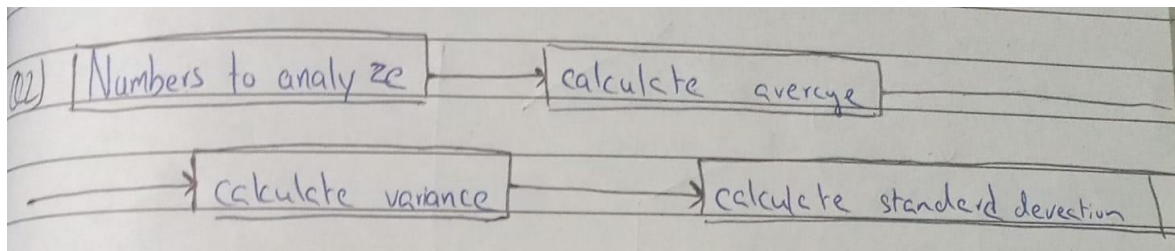
The function returns the standard deviation of a data set of n numbers  $x_0, \dots, x_{n-1}$ .

STDEV

```
#include <iostream>
#include <math.h>
using namespace std;

double stdev(double x[], int n) { // Declaring a function
    int size = n;
    double summation = 0;
    double sum = 0;
    for (int i = 0; i < size; i++) {
        sum += x[i] // Calculates Sum of the array
    }
    double avg = sum / size;
    for (int i = 0; i < size; i++) {
        sum += x[i]
        summation += pow(x[i] - avg, 2);
    }
    double stdv = sqrt(summation / size);
    return stdv;
}

int main() {
    double x[] = {1, 0, 0, 1};
    int size = sizeof(x) / sizeof(x[0]);
    double stdv = stdev(x, size);
    cout << "The standard deviation of the Array is: " << stdv;
}
```



```

\\CP\\CP_A2\\stdv\\\" ; if ($?) { g++ task.cpp -o task } ; if ($?) { .\\task }
The standard deviation of the Array is: 0.5

```

Q3) Write a program to multiply two 3x3 matrices. Make with function that can pass array to each function.

GetValue(PassArray)

ComputeAxA(PassArray)

PrintArray(PassArray)

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Matrix

```

#include <iostream>
using namespace std;

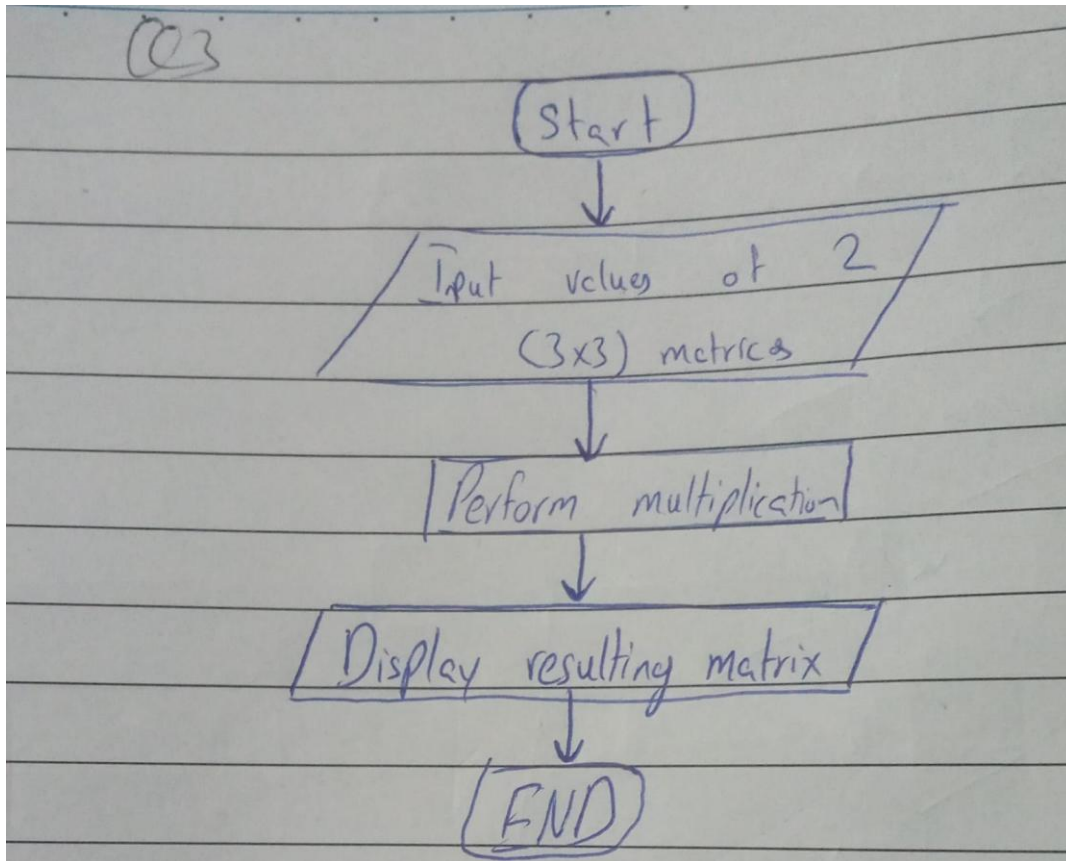
void GetValue(int matrix[3][3], string name){
    // Gets value of the matrices from the user
    for(int row = 0; row < 3; row++){
        for(int col = 0; col < 3; col++){
            cout << "Enter value for " << name << "[ " << row
                << "][ " << col << "]: ";
            cin >> matrix[row][col];
        }
    }
}

void ComputeAxB(int a[3][3], int b[3][3], int c[3][3]){
    // Multiplies the two matrices
    for(int i = 0; i < 3; i++){
        for(int j = 0; j < 3; j++){
            c[i][j] = 0;
            for(int k = 0; k < 3; k++){
                c[i][j] += a[i][k] * b[k][j];
            }
        }
    }
}

void PrintArray(int c[3][3]){
    // Prints any matrix
    for(int row = 0; row < 3; row++){
        for(int col = 0; col < 3; col++){
            cout << c[row][col] << " ";
        }
        cout << endl;
    }
}

```

Q3) Input 2 (3x3) matrices → Multiply and display the new (3x3) matrix



```
Enter value for Matrix A [0][0]: 1
Enter value for Matrix A [0][1]: 2
Enter value for Matrix A [0][2]: 2
Enter value for Matrix A [1][0]: 5
Enter value for Matrix A [1][1]: 6
Enter value for Matrix A [1][2]: 7
Enter value for Matrix A [2][0]: 5
Enter value for Matrix A [2][1]: 8
Enter value for Matrix A [2][2]: 9
Enter value for Matrix B [0][0]: 3
Enter value for Matrix B [0][1]: 2
Enter value for Matrix B [0][2]: 1
Enter value for Matrix B [1][0]: 1
Enter value for Matrix B [1][1]: 2
Enter value for Matrix B [1][2]: 3
Enter value for Matrix B [2][0]: 4
Enter value for Matrix B [2][1]: 3
Enter value for Matrix B [2][2]: 2
Printing the Computed array:
13    12    11
49    43    37
59    53    47
```

Q4) Write a program that takes a string from the user. The program then calculates the

- ✓ Total number of Characters
- ✓ Total number of Spaces
- ✓ Total number of Tabs
- ✓ Total number of Lines

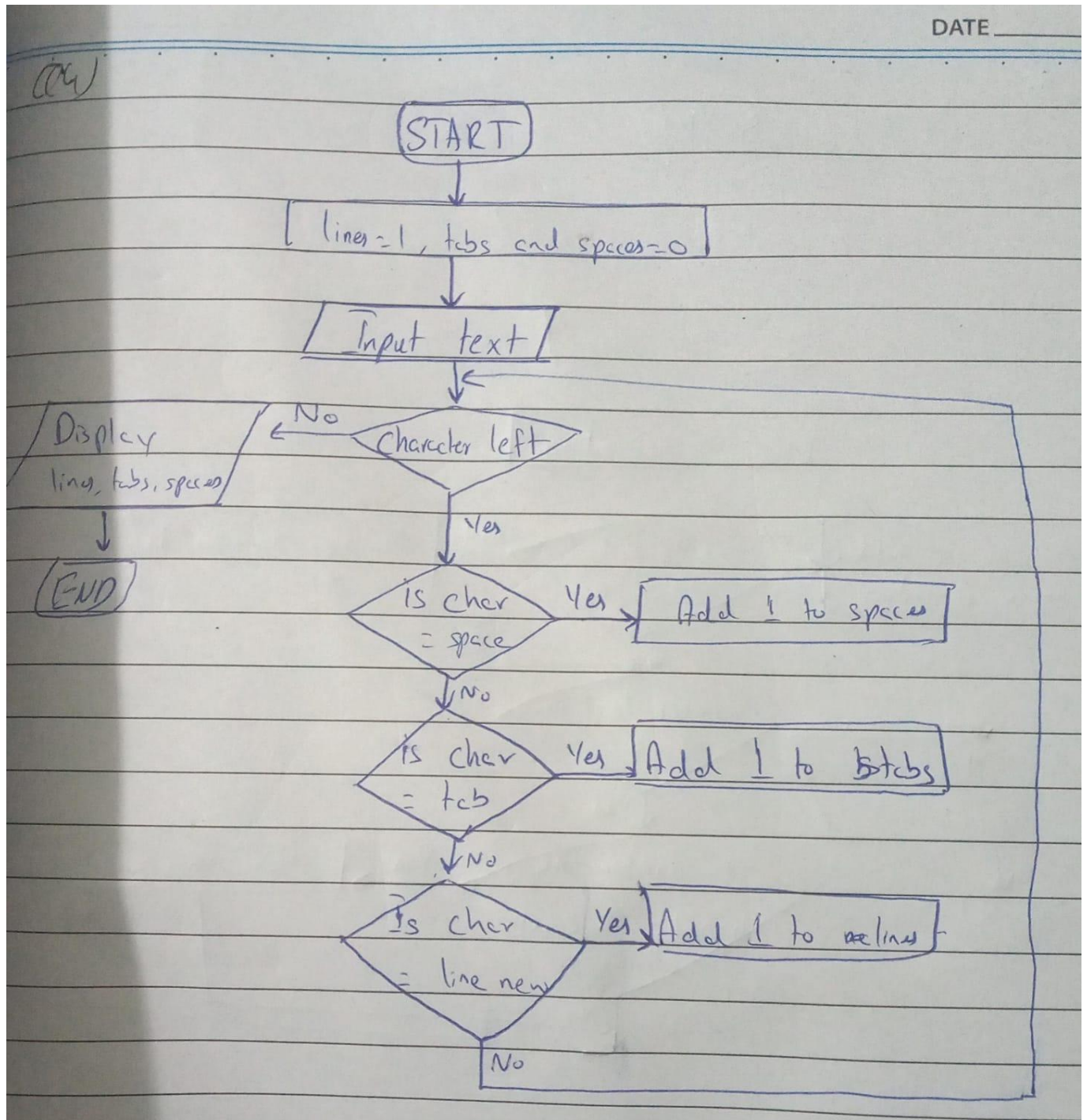
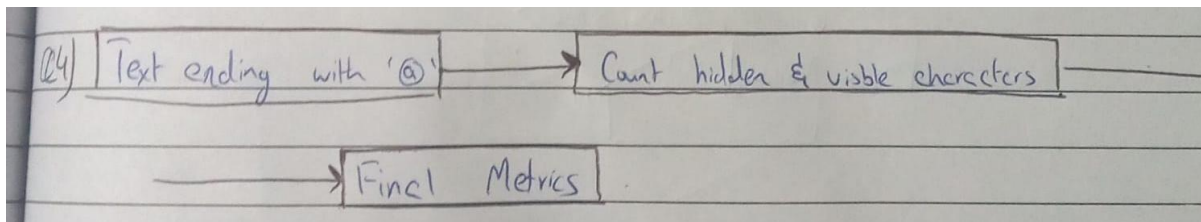
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### String Calculations

```
#include <iostream>
using namespace std;

void stringEval (int & tabs, int & spaces, int & lines, string sent) {
    // Iterate through the string character by character
    for (int i = 0; i < sent.size(); i++) {
        if (sent[i] == ' ')
            spaces++;
        else if (sent[i] == '\t')
            tabs++;
        else if (sent[i] == '\n')
            lines++;
    }
}

int main () {
    string sent;
    int tabs = 0, spaces = 0, lines = 1;
    cout << "Enter a string (Put '@' at the end of input and Press Enter):"
    << endl;
    getline (cin, sent, '@');
    // If the input is empty, reset the lines to 0
    if (sent.empty()) {
        lines = 0;
    }
    stringEval (tabs, spaces, lines, sent);
    cout << "Characters : " << sent.size() << endl << "spaces : "
    << spaces << endl << "Tabs : " << tabs << endl
    << "Lines : " << lines << endl;
    return 0;
}
```



```

Enter a string (Put '@' at the end of input and then Press Enter):
Hello This is falan
This is the second line @

=== String Metrics ===
Characters : 47
Spaces    : 9
Tabs      : 2
Lines     : 2

```

Q5) Write a program to calculate mean, median and mode of an array using function.

Stats

```

#include <iostream>
#include <algorithm>
using namespace std;

double getMean(int arr[], int n);
double getMedian(int arr[], int n);
int getMode(int arr[], int n);

int main() {
    int data[] = {4, 6, 7, 3, 8, 64, 55, 5, 7, 76, 8};
    int n = sizeof(data) / sizeof(data[0]);
    // Sorting for Median calculation
    sort(data, data + n);
    cout << "Sorted Array: ";
    for (int i = 0; i < n; i++)
        cout << data[i] << " ";
    cout << "\n\n      Statics      " << endl;
    cout << "Mean : " << getMean(data, n) << endl;
    cout << "Median : " << getMedian(data, n) << endl;
    cout << "Mode : " << getMode(data, n) << endl;
    return 0; }

```

```
double getMean (int arr[], int n) {
```

```
    double sum = 0;
```

```
    for (int i = 0; i < n; i++)
```

```
        sum += arr[i];
```

```
    return sum/n;
```

```
}
```

```
double getMedian (int arr[], int n) {
```

```
    if (n % 2 != 0)
```

```
        return arr[n/2]
```

```
    } else {
```

```
        return (arr[(n-1)/2] + arr[n/2]) / 2.0;
```

```
    } }
```

```
int getMode (int arr[], int n) {
```

```
    int mode = arr[0];
```

```
    int max_count = 1;
```

```
    int current_count = 1;
```

```
    for (int i = 1; i < n; i++) {
```

```
        if (arr[i] == arr[i-1]) {
```

```
            current_count ++;
```

```
        } else {
```

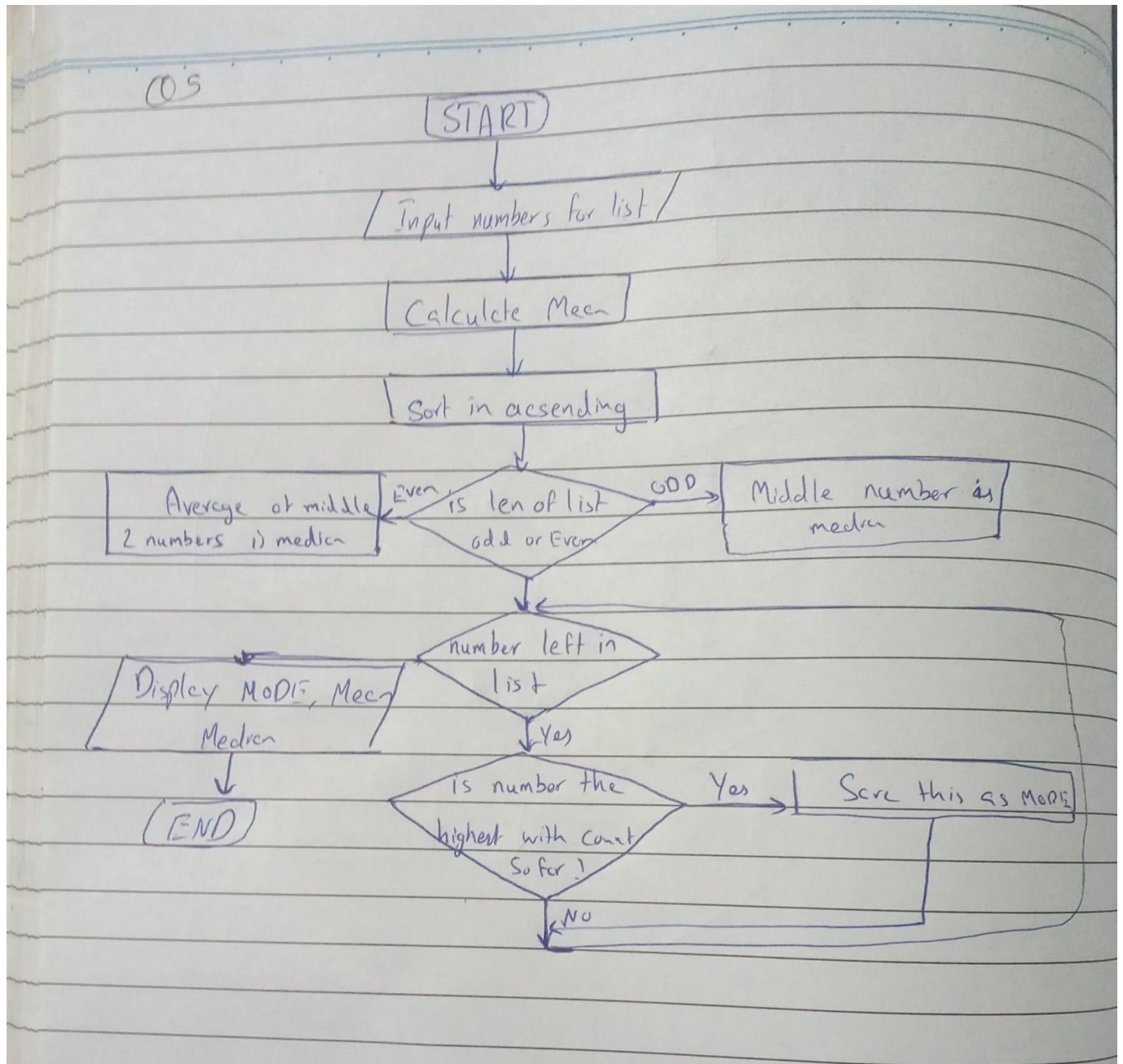
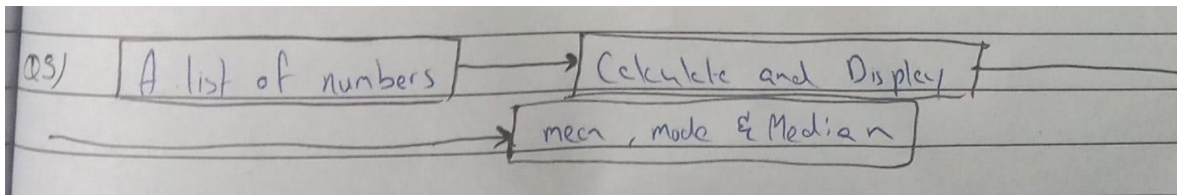
```
            current_count = 1;
```

```
        }
```

```
        if (current_count > max_count) {
```

```
            max_count = current_count; mode = arr[i-1]; }
```

```
    return mode; }
```



Sorted Array: 3 3 3 4 4 4 5 5 5 6 6 6 7 7 8 8 8 25 63 64 64 76 95

=====Statistics

Mean : 20.8261

Median : 6

Mode : 3